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原创研究文章

基于真实世界研究的机器学习模型，用于预测由中药制剂（正清风痛宁）引起的类风湿关节炎患者异常肝功能



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摘要

目的：类风湿关节炎（RA）是一种影响全身小关节的系统性自身免疫性疾病，会降低患者的生活质量。正清风痛宁（ZF）是一种用于治疗RA的传统中药制剂。ZF可能引起肝损伤。本研究的目的是建立一个预测ZF引起的肝功能异常的模型。

方法：本回顾性研究收集了2018年1月至2023年4月间多中心的数据。根据丙氨酸转氨酶（ALT）水平，将肝功能异常设为目标变量。通过单变量分析和顺序前向选择对特征进行筛选以进行建模。比较了十种机器学习和深度学习模型，以找到最有效利用现有数据预测肝功能的模型。

结果：本研究纳入了1,913名符合条件的患者。在10种学习模型中，LightGBM模型表现最佳（准确率 = 0.96）。LightGBM模型的预测指标如下：预测ALT < 40 U/L的精确度为 = 0.99，召回率为 = 0.97，F1_分数为 = 0.98，曲线下面积（AUC）为 = 0.98，敏感性为 = 0.97，特异性为 = 0.85；预测40 ALT < 80 U/L的精确度为 = 0.60，召回率为 = 0.83，F1_分数为 = 0.70，AUC为 = 0.98，敏感性为 = 0.83，特异性为 = 0.97；预测ALT 80 U/L的精确度为 = 0.83，召回率为 = 0.63，F1_分数为 = 0.71，AUC为 = 0.97，敏感性为 = 0.63，特异性为 = 1.00。ZF引起的肝功能异常与总胆固醇和甘油三酯水平升高、TNF-α抑制剂、JAK抑制剂、甲氨蝶呤 + 非甾体抗炎药、来氟米特、吸烟、年龄较大及 45–65 岁中年女性相关。

结论：本研究开发了一个用于预测ZF引起的肝功能异常的模型，这可能有助于提高ZF与西药联合使用的安全性。

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1. 介绍

类风湿性关节炎（RA）是一种系统性自身免疫性疾病，常影响全身的小关节，导致关节疼痛、压痛、肿胀和早晨僵硬以及其他症状^[1,2]。RA 的发病率是最常见的疾病之一

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Original Research Article

A machine learning model for predicting abnormal liver function induced by a Chinese herbal medicine preparation (Zhengqing Fengtongning) in patients with rheumatoid arthritis based on real-world study



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ABSTRACT

Objective: Rheumatoid arthritis (RA) is a systemic autoimmune disease that affects the small joints of the whole body and degrades the patients' quality of life. Zhengqing Fengtongning (ZF) is a traditional Chinese medicine preparation used to treat RA. ZF may cause liver injury. In this study, we aimed to develop a prediction model for abnormal liver function caused by ZF.

Methods: This retrospective study collected data from multiple centers from January 2018 to April 2023. Abnormal liver function was set as the target variable according to the alanine transaminase (ALT) level. Features were screened through univariate analysis and sequential forward selection for modeling. Ten machine learning and deep learning models were compared to find the model that most effectively predicted liver function from the available data.

Results: This study included 1,913 eligible patients. The LightGBM model exhibited the best performance (accuracy = 0.96) out of the 10 learning models. The predictive metrics of the LightGBM model were as follows: precision = 0.99, recall rate = 0.97, F1_score = 0.98, area under the curve (AUC) = 0.98, sensitivity = 0.97 and specificity = 0.85 for predicting ALT < 40 U/L; precision = 0.60, recall rate = 0.83, F1_score = 0.70, AUC = 0.98, sensitivity = 0.83 and specificity = 0.97 for predicting $40 \leq \text{ALT} < 80$ U/L; and precision = 0.83, recall rate = 0.63, F1_score = 0.71, AUC = 0.97, sensitivity = 0.63 and specificity = 1.00 for predicting ALT ≥ 80 U/L. ZF-induced abnormal liver function was found to be associated with high total cholesterol and triglyceride levels, the combination of TNF-α inhibitors, JAK inhibitors, methotrexate + nonsteroidal anti-inflammatory drugs, leflunomide, smoking, older age, and females in middle-age (45–65 years old).

Conclusion: This study developed a model for predicting ZF-induced abnormal liver function, which may help improve the safety of integrated administration of ZF and Western medicine.

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1. Introduction

Rheumatoid arthritis (RA) is a systemic autoimmune disease that often affects small joints throughout the body, leading to joint pain, tenderness, swelling and morning stiffness, and other symptoms ^[1,2]. The incidence of RA, which is one of the most common

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炎性关节炎类型在中国估计约为0.4% [3]。正清风痛宁（ZF）是一种现代中药制剂，由中药青风藤（河南 金樱子属 [Thunb.] Rehd. et Wils. var. cinereum Rehd. et Wils) [4]中提取的生物碱精制而成。西诺敏是ZF的主要活性成分，常用于中国与西药联合治疗类风湿性关节炎（RA）及其他风湿性疾病，如甲氨蝶呤（MTX）、伊古替莫德（IGU）和来氟米特（LEF） [5]。RA是一种慢性疾病，患者通常需要长期治疗，这可能引发一些不良药物反应（ADRs）。

由思诺敏引起的常见不良反应包括腹胀、腹泻、白细胞计数减少，以及全身或局部皮疹[6,7]。肝功能异常是一种罕见但可能严重的不良反应，可能导致危及生命的后果。目前，很少有研究探讨使用ZF的患者肝功能异常的预测。药物的不良反应可能受到多种因素的影响，如性别、年龄和药物联合使用，并且不同患者的不良反应严重程度可能不同[8]。因此，有必要在个体化水平预测患者的肝功能异常。

在当前的医学研究中，来自临床实践的真实世界证据（RWE）是一个重要的数据来源[9]。RWE可以从电子病历、实验室数据以及诊断和治疗期间或之后的随访记录中收集。与临床试验相比，基于RWE的研究与临床实践更相关，并且可以应用于个体化方案，具有丰富的证据和良好的外部有效性[10,11]。机器学习和深度学习算法，如极端梯度提升（XGBoost）、类别提升（CatBoost）、轻量级梯度提升机（LightGBM）、梯度提升决策树（GBDT）和TabNet[12–16],, 可以用于解析来自RWE的复杂多维数据。这些算法擅长数据挖掘、特征提取和模型开发。通过处理大量数据，可以从多个变量中评估数据驱动的估计，并能识别非线性变量关系，从而实现高精度的临床预测

本研究旨在建立一个来自多中心的类风湿性关节炎真实世界数据库，以探讨与ZF给药后肝功能异常相关的重要因素，并构建由ZF引起肝功能异常的预测模型。对于高风险的肝功能异常患者，通过及时且有效地调整用药方案，可以降低严重结局的发生率。

2. 方法

2.1. 研究设计与人群

这是一项回顾性研究。我们在中国的多家三级甲等医院招募了患者，包括苏州大学附属第一医院、南方医科大学南方医院

从2018年1月到2023年4月，我们在河北医科大学及河北省人民医院招募了患者。我们还在海军军医大学第二附属医院（上海长征医院）从2022年6月1日到2023年5月31日招募患者进行外部验证。目标变量是丙氨酸转氨酶（ALT）水平。根据《不良事件通用术语标准》（CTCAE v4.0）量表以及医生的实际临床经验， $ALT \leq < 40$ U/L被认为表示肝功能正常，而 $40 < ALT \leq < 80$ U/L被认为表示肝功能异常， $ALT \geq 80$ U/L被认为表示肝功能严重异常 [22]。基于临床知识和既往文献，我们选择了与ZF诱导的肝功能异常相关的38个特征进行筛选，并随后用于开发ZF诱导的肝功能异常预测模型。

纳入标准如下：(1) 诊断为类风湿性关节炎（RA）的患者；(2) 年龄在18岁及以上的患者；(3) 正在接受RA治疗的患者；(4) 门诊或住院患者，且随访记录至少12周。RA治疗包括：甲氨蝶呤（MTX）、伊曲康肽（IGU）、来氟米特（LEF）、MTX + ZF、IGU + ZF、LEF + ZF、MTX + LEF、MTX + 非甾体抗炎药（NSAIDs）、MTX + 总苕药苷、MTX + 昆完胶囊、MTX + 必奇胶囊、MTX + LEF +ZF、MTX + NSAIDs + ZF、Janus 激酶（JAK）抑制剂 + ZF、JAK抑制剂 + MTX、JAK抑制剂 + IGU、肿瘤坏死因子- α （TNF- α ）抑制剂 + ZF、TNF- α 抑制剂 + MTX，以及TNF- α 抑制剂 + IGU。排除标准如下：(1) 患者患有其他风湿性疾病；(2) 患者正在妊娠或哺乳期；(3) 患者未使用ZF；(4) 患者缺乏主要肝功能数据或其他主要研究数据。

本研究获得了苏州大学第一附属医院（编号2023-268）医学伦理委员会的批准，南方医科大学南方医院（编号）国家国家电力公司（NFEC-2023-438）和河北总医院（编号2023-324）。该研究按照1964年赫尔辛基宣言进行，研究数据被完全去识别，并根据国际医学科学组织理事会/世界卫生组织《涉及人类健康研究的国际伦理指南》（2016）[23]，删除了患者的保密信息。由于这是一项基于电子病历数据的大型回顾性研究，且患者信息已完全去识别，所有伦理委员会均批准了知情同意豁免。

2.2. 数据收集与处理

从患者的电子病历中收集了以下数据：人口学信息（如年龄和性别）、每日ZF剂量、吸烟状况、有无心血管疾病（CVD）、药物组合、检测指标（如红细胞沉降率（ESR）、C反应蛋白（CRP）、类风湿因子（RF）、甘油三酯（TG）和总胆固醇（TC））以及基于CRP水平的28关节疾病活动评分。

缺失数据率超过50%的变量被剔除。缺失数据率是衡量数据集中缺失比例的比例的指标。样本量极度不平衡的变量，其正样本与总样本数的比例小于0.1，也被删除。分类变量如性别、吸烟和心血管疾病（CVD）通过独热编码（one-hot encoding）编码，而性别/年龄组、ZF剂量及药物组合则按照下表（表1）采用序数编码（ordinal encoding）编码。

2.3. 特征选择

首先，对所有与ZF引起的异常肝功能相关的38个特征进行了单变量分析。显著性通过Kruskal-Wallis检验或卡方检验进行测试，那些

inflammatory arthritis types, is estimated to be approximately 0.4% in China [3]. Zhengqing Fengtongning (ZF) is a modern traditional Chinese medicine preparation refined from the alkaloids extracted from the Chinese herbal medicine Qingfengteng (*Sinomenium acutum* [Thunb.] Rehd. et Wils. var. cinereum Rehd. et Wils) [4]. Sinomenine is the main active ingredient of ZF and is commonly used to treat RA and other rheumatic diseases in China in combination with Western medicine, such as methotrexate (MTX), iguratimod (IGU) and leflunomide (LEF) [5]. RA is a chronic disease that patients generally require long-term treatment, which may induce some adverse drug reactions (ADRs).

The common ADRs caused by sinomenine include bloating, diarrhea, low white blood cell count, and whole body or local rash [6,7]. Abnormal liver function is a rare but potentially severe ADR that can lead to life-threatening consequences. Currently, few studies have explored the prediction of abnormal liver function in patients using ZF. Adverse reactions to a drug can be affected by many factors, such as gender, age and drug combination, and the severity of ADRs may vary in different patients [8]. Therefore, it is necessary to predict abnormal liver function in patients at individualized levels.

In current medical research, real-world evidence (RWE) from clinical practice is an important data source [9]. RWE can be collected from electronic medical records, laboratory data, and follow-up records during and/or after diagnosis and treatment. Compared with clinical trials, studies based on RWE are more relevant to clinical practice and can be applied in individualized programs with rich evidence and good external validity [10,11]. Machine learning and deep learning algorithms, such as extreme gradient boosting (XGBoost), categorical boosting (CatBoost), light gradient boosting machine (LightGBM), gradient boosting decision tree (GBDT), and TabNet [12–16], can be used to interpret sophisticated and multidimensional data from RWE. These algorithms are adept at data mining, feature extraction, and model development. By processing large amounts of data, data-driven estimates can be evaluated from multiple variables, and nonlinear variable relationships can be identified, enabling high-precision prediction of clinical outcomes [17,18]. The use of machine learning and deep learning in clinical medicine research has seen increasing interest in recent years. For instance, a GBDT model was shown to have higher precision than global trigger tools for predicting adverse drug events in 1,746 pediatric patients (44% vs 13.3%) [19]; an ensemble model using five machine learning algorithms achieved greater R^2 in predicting vancomycin concentration, compared with the population pharmacokinetic model (0.6 vs 0.3) [20]; and a XGBoost model was able to construct a personalized medicine model of Shengmai injection [21]. With the larger sample size of input data, machine learning and deep learning models can be continually optimized to achieve better performance and practicality. We believe that taking machine learning and deep learning-based models from development to real-world application is worth further research.

This study aimed to develop a real-world database of RA from multiple centers to explore the important factors that are linked to abnormal liver function following administration of ZF and to construct a predictive model for abnormal liver function caused by ZF. For patients at high risk of abnormal liver function, the incidence of severe outcomes could be reduced by timely and effective adjustment of the medication regimen.

2. Methods

2.1. Study design and population

This was a retrospective study. We enrolled patients at multiple Grade-A tertiary hospitals in China, including the First Affiliated Hospital of Soochow University, Nanfang Hospital of Southern

Medical University, and Hebei General Hospital, from January 2018 to April 2023. We also enrolled patients at the Second Affiliated Hospital of Naval Medical University (Shanghai Changzheng Hospital) from June 1, 2022 to May 31, 2023 for external validation. The target variable was the alanine transaminase (ALT) level. According to the Common Terminology Criteria for Adverse Events (CTCAE v 4.0) scale and practical clinical experience of the physicians, $ALT < 40$ U/L was considered to indicate normal liver function, while $40 \leq ALT < 80$ U/L was considered to indicate abnormal liver function, and $ALT \geq 80$ U/L was considered to indicate severely abnormal liver function [22]. A total of 38 features associated with ZF-induced abnormal liver function were chosen based on clinical knowledge and previous literature for screening and were subsequently used to develop a prediction model of ZF-induced abnormal liver function.

The inclusion criteria were as follows: (1) patients diagnosed with RA; (2) patients aged ≥ 18 years; (3) patients receiving treatment for RA; and (4) outpatients or inpatients with at least 12 weeks of follow-up records. RA treatments include: MTX, IGU, LEF, MTX + ZF, IGU + ZF, LEF + ZF, MTX + LEF, MTX + nonsteroidal anti-inflammatory drugs (NSAIDs), MTX + total glucosides of paeony, MTX + Kunxian Capsule, MTX + Biqi Capsule, MTX + LEF + ZF, MTX + NSAIDs + ZF, Janus kinase (JAK) inhibitor + ZF, JAK inhibitor + MTX, JAK inhibitor + IGU, tumor necrosis factor- α (TNF- α) inhibitor + ZF, TNF- α inhibitor + MTX, and TNF- α inhibitor + IGU. The exclusion criteria were as follows: (1) patients had other rheumatic diseases; (2) patients were pregnant or breastfeeding; (3) patients did not use ZF; and (4) patients lacked primary liver function data or other primary research data.

This study was approved by the medical ethics committees of the First Affiliated Hospital of Soochow University (No. 2023-268), Nanfang Hospital of Southern Medical University (No. NFEC-2023-438) and Hebei General Hospital (No. 2023-324). The study was conducted in accordance with the *Helsinki Declaration* of 1964, study data were fully deidentified, and confidential information was deleted from the patients in accordance with the Council for International Organizations of Medical Sciences/World Health Organization *International Ethical Guidelines for Health-related Research Involving Humans* (2016) [23]. As this was a large retrospective study based on electronic medical record data and patient information was fully deidentified, informed consent exemptions were approved by all the ethics committees.

2.2. Data collection and processing

The following data were collected from the patients' electronic medical records: demographic information (such as age and gender), daily ZF dose, smoking status, presence or absence of cardiovascular disease (CVD), drug combinations, assay indices (such as erythrocyte sedimentation rate [ESR], C-reactive protein [CRP], rheumatoid factor [RF], triglyceride [TG], and total cholesterol [TC]), and the disease activity score in 28 joints based on CRP level.

Variables with a missing data rate greater than 50% were eliminated. The missing data rate is a measure of the proportion of missing values in a dataset. The variables with highly imbalanced sample sizes, characterized by a ratio of positive samples to the total number of samples less than 0.1, were deleted. The classification variables such as gender, smoking and CVD were coded by one-hot encoding, and gender/age groups, ZF doses, and drug combinations were coded by ordinal encoding as follows (shown in Table 1).

2.3. Feature selection

First, a univariate analysis was conducted for all 38 features associated with ZF-induced abnormal liver function. Significance was tested by Kruskal-Wallis test or chi-square test, and those

表 1 变量编码。

Item	通过序数编码编码						
	0	1	2	3	4	5	6
性别	Male	女性					
年龄/性别	年轻男性	年轻女性	中年	中年	年长男性	年长女性	
群组			male	女性			
ZF日剂量	60 毫克	120 毫克	180 毫克	240 毫克			
Drug	MTX +ZF	IGU +ZF	LEF + MTX + ZF	LEF +ZF	MTX + 非甾体抗炎药 + ZF	TNF-α 抑制剂 + ZF JAK 抑制剂 + ZF	
组合							
丙氨酸转氨酶水平	丙氨酸转氨酶 < 40 U/L	40 单位/升 ≤ ALT < 80 U/L	ALT ≥ 80 单位/升				

ALT: 丙氨酸转氨酶; IGU: 依鲁替摩; JAK: Janus激酶; LEF: 来氟米特; MTX: 甲氨蝶呤; NSAID: 非甾体抗炎药; TNF: 肿瘤坏死因子; ZF: 正清风痛宁。

P 值为 < 0.05 的被认为是导致 ZF 诱导的肝功能异常的重要变量。随后, 基于 XGBoost 的顺序前向选择 (SFS) 算法被应用于进一步筛选 [24]。选择过程涉及通过 SFS 算法每次向特征子集中添加一个特征, 迭代生成新模型, 并计算模型性能。模型性能用 F1_ 分数表示, 该分数是精确率和召回率的综合评价指标, 其中 F1_ 分数越高表示模型越稳健。当使用某一特征子集构建的模型 F1_ 分数达到最高值时, 迭代停止。重要变量被纳入该具有最小变量集和最佳性能的特征子集中。

2.4. 模型开发与解读

如图1所示, scikit-learn库中的train_test_split函数被用来以4:1的比例随机将数据集划分为训练集和测试集。为了生成一个无偏且可靠的预测模型, 进行了6折交叉验证。

在训练集中进行了导管, 并通过参数调优以F1_分数作为评估指标来优化模型。针对训练集, 利用10种算法开发了ZF诱导的异常肝功能预测模型, 并比较了它们的预测性能。这些算法包括逻辑回归、随机森林、人工神经网络、支持向量机、k近邻、CatBoost、LightGBM、XGBoost、GBDT和TabNet, 它们作为常见算法类型具有良好的预测能力。十种机器学习和深度学习模型的发展原理显示在补充文件1中。算法的预测性能在测试集上进行了评估, 并选择性能最佳的算法来开发最终的预测模型。10个模型的参数显示在表S1中。

随后, 通过预测性能最好的算法计算了所选变量的重要性得分。变量的重要性是指模型中每个变量在提高整个模型预测能力方面的贡献程度。随后, 应用Shapley加性解释 (SHAP) 来解释重要变量的影响。

Table 1
Variable encoding.

Item	Coded by ordinal encoding						
	0	1	2	3	4	5	6
Gender	Male	Female					
Age/gender group	Young male	Young female	Middle-aged male	Middle-aged female	Elderly male	Elderly female	
ZF daily dose	60 mg	120 mg	180 mg	240 mg			
Drug combination	MTX + ZF	IGU + ZF	LEF + MTX + ZF	LEF + ZF	MTX + NSAIDs + ZF	TNF- α inhibitor + ZF	JAK inhibitor + ZF
ALT level	ALT < 40 U/L	40 U/L $\leq$ ALT < 80 U/L	ALT $\geq$ 80 U/L				

ALT: alanine transaminase; IGU: iguratimod; JAK: Janus kinase; LEF: leoflunomide; MTX: methotrexate; NSAID: nonsteroidal anti-inflammatory drug; TNF: tumor necrosis factor; ZF: Zhengqing Fengtongning.

with a P value < 0.05 were considered important variables to ZF-induced abnormal liver function. The sequential forward selection (SFS) algorithm based on XGBoost was subsequently applied for further screening [24]. The selection process involved adding one feature to the feature subset each time via the SFS algorithm, iteratively generating a new model, and calculating the model performance. The model performance was expressed by the F1_score, which is a comprehensive evaluation index of precision and recall rate, where a higher F1_score indicates a more robust model. Iteration stopped when the F1_score of the model built with a certain feature subset reached the highest value. Important variables were included in this feature subset with the minimum variable set and optimum performance.

2.4. Model development and interpretation

As shown in Fig. 1, the train_test_split function in the scikit-learn library was used to randomly divide the dataset into a training set and a testing set at a ratio of 4:1. To generate an unbiased and reliable prediction model, a 6-fold cross-validation was con-

ducted in the training set, and the model was optimized by parameter tuning with the F1_score as the evaluation index. With respect to the training set, prediction models for ZF-induced abnormal liver function were developed by 10 algorithms, and their predictive performance was compared. These algorithms, which include logistic regression, random forest, artificial neural network, support vector machine, k-nearest neighbor, CatBoost, LightGBM, XGBoost, GBDT and TabNet, have good predictive ability as common algorithm types. The development principles of the ten machine learning and deep learning models are shown in Supplementary File 1. The predictive performance of the algorithms was evaluated on the testing set, and the algorithm with the best performance was selected to develop the final prediction model. The parameters of the 10 models are displayed in Table S1.

Subsequently, importance scores for the selected variables were calculated by the algorithm with the best predictive performance. The importance of a variable refers to the degree to which each variable in the model contributes to improving the predictive power of the whole model. Afterward, Shapley additive explanation (SHAP) was applied to interpret the impacts of important vari-

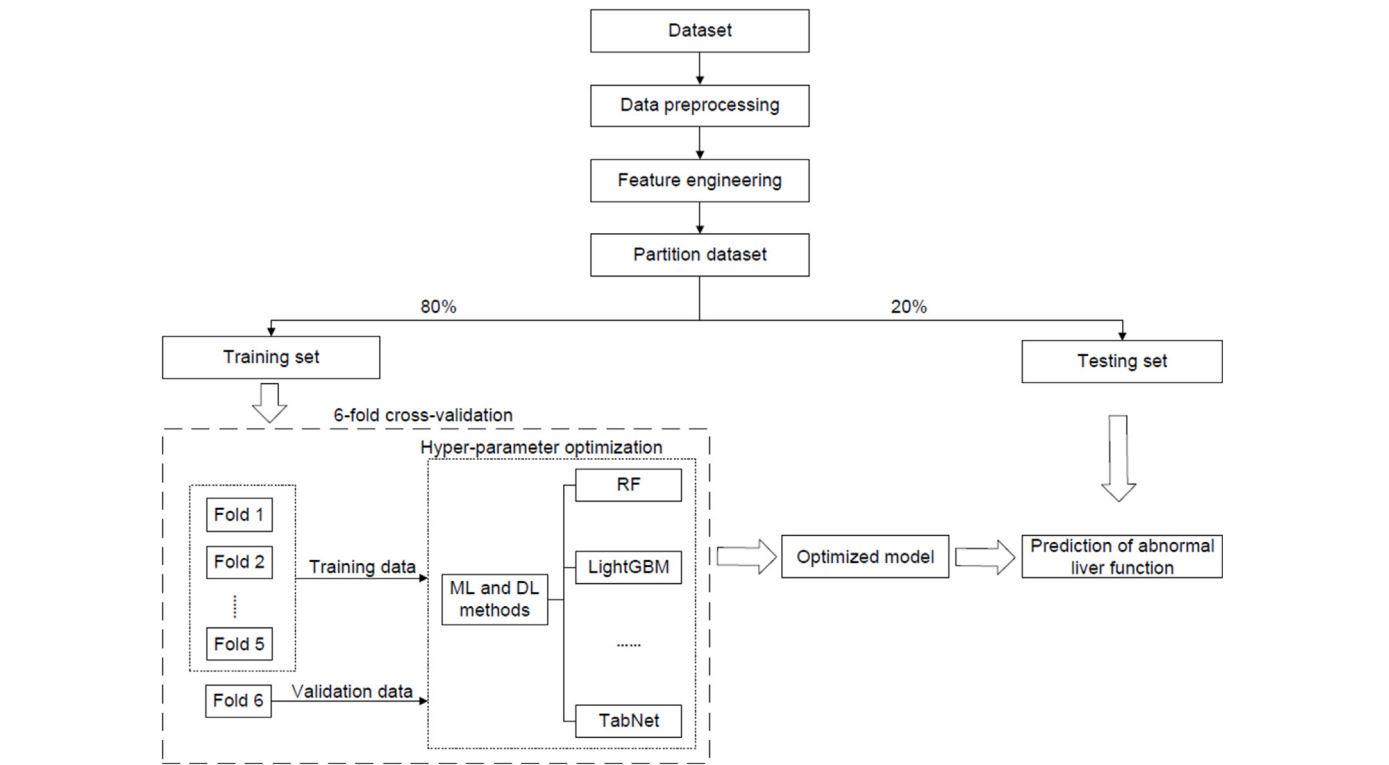


图1. 数据处理工作流程 归一化和建模。DL: 深度学习; LightGBM: 轻量级梯度提升机; ML: 机器学习; RF: 随机森林

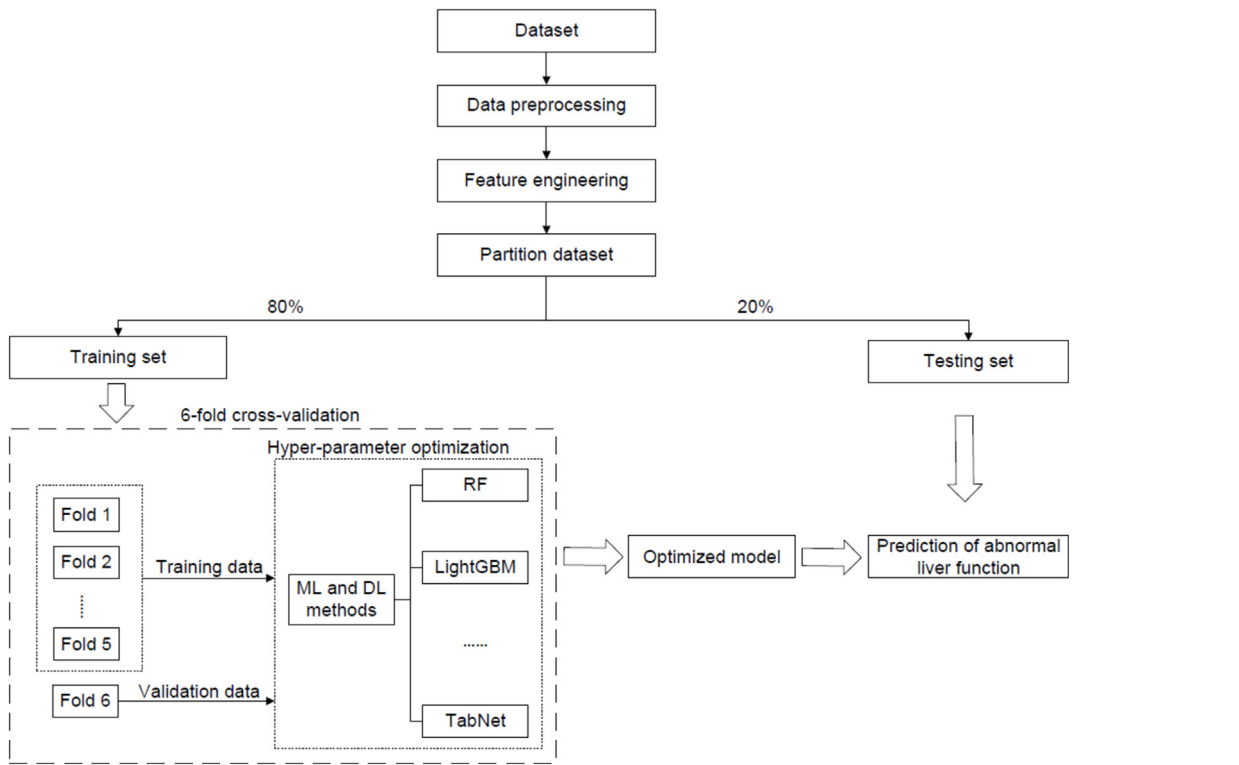


Fig. 1. Workflow of data processing and modeling. DL: deep learning; LightGBM: light gradient boosting machine; ML: machine learning; RF: random forest.

在模型输出 [25] 上进行了分析。最终，使用混淆矩阵来计算精确率和召回率，并在测试集中可视化预测模型的性能。

2.5. 外部验证和模型应用

使用混淆矩阵来显示外部验证的预测性能。最终，为了方便模型的应用，我们建立了一个网络应用程序，可以为个体患者提供ZF诱发异常肝功能预测的决策支持。

2.6. 统计分析

在单变量分析中，连续变量使用 SciPy 库（版本 1.7.3）通过 Kruskal-Wallis 检验进行分析。无序分类变量通过卡方检验进行分析。所有统计分析均使用双侧检验，并报告检验统计量及相应的 P 值。P 值 < 0.05 被认为具有统计学意义。

为了评估模型的预测性能，计算了包括精确率、召回率、F1_值、准确率、AUC、敏感性和特异性在内的评估指标。F1_值用于衡量模型的优缺点，F1_值越高表示模型性能越好。评估指标的具体公式如下：

$$\begin{aligned} \text{精确率} &= \text{TP} / (\text{TP} + \text{FP}). \quad \text{召回率} = \text{TP} / (\text{TP} + \text{FN}). \quad \text{F1_分数} = 2 \text{ TP} / (2 \text{ TP} + \text{FP} + \text{FN}). \\ \text{准确率} &= (\text{TP} + \text{TN}) / (\text{TP} + \text{FN} + \text{FP} + \text{TN}). \quad \text{敏感性} = \text{TP} / (\text{TP} + \text{FN}). \quad \text{特异性} = \text{TN} / (\text{TN} + \text{FP}). \end{aligned}$$

TP：真正例，表示被正确识别为阳性的病例数量；TN：真负例，表示被正确识别为阴性的病例数量；FP：假阳性，表示被错误识别为阳性的阴性病例数量；FN：假阴性，表示被错误识别为阴性的阳性病例数量。

使用 Matplotlib（版本 3.8.2）软件包构建了混淆矩阵。所有机器学习 and 深度学习算法实验均在 Windows 10 上运行，配置为 Intel Core i5-10400F CPU @ 2.90 GHz 12 核 CPU 和 512 GB 内存。数据分析使用 Python 3.9.12 和 IBM SPSS Statistics 22 进行。

3. 结果

3.1. 基线信息

患者纳入的工作流程如图 2 所示。共识别出 65,007 例类风湿性关节炎（RA）患者。其中，56,701 例患者年龄为 18 岁。在这些患者中，46,138 例接受过 RA 药物治疗，其中 4,017 例患者有至少 12 周的随访记录。其他风湿性疾病患者（34 例）、孕妇或哺乳期妇女（9 例）、未使用 ZF 的患者（1,964 例）或缺少主要研究数据的患者（97 例）被排除。最终，本研究共纳入 1,913 例患者。其中，收集自苏州大学附属第一医院的患者分别为 916 例、692 例和 305 例。

分别是南方医科大学附属医院和河北省人民医院。

根据4:1的比例，培训集包含1530名患者，测试集包含383名患者。数据集中患者的人口统计学和临床特征显示在表2中。男性有1371名（71.67%），女性有542名（28.33%）。整个数据集个体的中位年龄为58岁（四分位数范围 [IQR] 50–68岁）。在性别_年龄组中，中年男性所占比例最高（37.11%），年轻女性所占比例最低（4.81%）。ZF的日剂量分为四个亚组：60 mg（n [%] = 108 [5.65%]）、120 mg（n [%] = 900[47.05%], 180）、[%] = 192 [10.04%]mg，以及240 mg（n [%] = 713[37.27%]）。有599名（31.31%）患者吸烟，371名（19.39%）患有心血管疾病（CVD）。在治疗方案中，使用MTX + ZF、IGU + ZF、LEF + MTX + ZF、LEF + ZF、MTX + NSAIDs + ZF、TNF-α抑制剂 + ZF以及JAK抑制剂 + ZF治疗的患者人数（%）为468（24.46%）。

共有176名肝功能异常的患者，他们的人口学和临床特征显示在表S2中。其中，男性126人，女性50人。中位年龄为68.5岁（四分位数间距56–75岁）。在性别_年龄组中，老年男性所占比例最高（56.25%）。在日剂量子组中，120 mg的剂量是最常用的（51.70%）。约49%的患者吸烟，12.50%患有心血管疾病。MTX ZF是最常用的治疗方案（31.25%）。

3.2. 变量选择

单变量分析的结果显示在表3中。共有22个显著变量被选出，P < 0.05，包括年龄、性别_年龄、心血管疾病状态、治疗方案、ZF日剂量、C反应蛋白、类风湿因子、红细胞沉降率、血红蛋白、前白蛋白、尿酸、总蛋白、白蛋白、谷氨酰转肽酶、中性粒细胞绝对值、平均血小板体积、平均红细胞体积、平均血红蛋白浓度、平均血红蛋白、红细胞压积、总胆固醇和甘油三酯。

之后，这22个变量通过SFS方法进行了进一步筛选。基于使用1到22个变量开发的XGBoost模型，获得了每个XGBoost模型的F1_分数（图3）。随着包含变量数量的增加，模型的F1_分数持续上升，并在模型包含6个变量时达到最大值（F1_分数 = 0.73）。由于追求简洁且准确的模型，采用了能够实现良好模型性能的最小变量集。前6个变量是性别_年龄、治疗方案、年龄、甘油三酯、吸烟状况和总胆固醇。

基于22个和6个变量的不同模型的F1_分数如图4和表S3所示。根据图4A，在基于单变量分析获得的22个变量的模型中，LightGBM模型显示出最佳结果，其平均F1_分数为0.77，标准差（Std）为0.07。在图4B中，对于基于SFS获得的6个变量的模型，LightGBM模型同样取得最佳结果，平均F1_分数为0.78，Std为0.07。比较使用6个变量和22个变量开发的模型发现，使用6个变量开发的模型具有更高的F1_分数，既简洁又稳健。因此，在测试集中使用这6个变量进行建模。训练集和测试集之间的目标变量及重要影响变量的差异如表S4所示。可以看出，两组数据集之间的变量没有显著差异（P值 > 0.05）。

ables on the model output [25]. Eventually, a confusion matrix was used to calculate the precision and recall rate and to visualize the performance of the prediction model in the testing set.

2.5. External validation and model application

A confusion matrix was used to show the predictive performance of the external validation. Eventually, in order to facilitate the application of the model, we established a web application that can provide decision support for generating predictions of ZF-induced abnormal liver function for individual patients.

2.6. Statistical analysis

In the univariate analysis, continuous variables were analyzed using the Kruskal-Wallis test via SciPy library (version 1.7.3). Disordered categorical variables were analyzed by the chi-square test. All the statistical analyses were conducted using two-sided tests, and the test statistics and corresponding *P* values are reported. A *P* value < 0.05 was considered to indicate statistical significance.

To evaluate the predictive performance of the models, evaluation metrics, including precision, recall rate, F1_score, accuracy, AUC, sensitivity and specificity, were calculated. The F1_score is used to measure the merits and defects of the model, and a higher F1_score indicates better model performance. The specific formulae for the evaluation metrics are as follows:

$$\begin{aligned} \text{Precision} &= \text{TP} / (\text{TP} + \text{FP}). \\ \text{Recall rate} &= \text{TP} / (\text{TP} + \text{FN}). \\ \text{F1_score} &= 2 \times \text{TP} / (2 \times \text{TP} + \text{FP} + \text{FN}). \\ \text{Accuracy} &= (\text{TP} + \text{TN}) / (\text{TP} + \text{FN} + \text{FP} + \text{TN}). \\ \text{Sensitivity} &= \text{TP} / (\text{TP} + \text{FN}). \\ \text{Specificity} &= \text{TN} / (\text{TN} + \text{FP}). \end{aligned}$$

TP: true positive, indicating the number of positive cases that were correctly identified; TN: true negative, indicating the number of negative cases that were correctly identified; FP: false positive, indicating the number of negative cases identified as positive; FN: false negative, indicating the number of positive cases identified as negative.

A confusion matrix was constructed with the Matplotlib (version 3.8.2) package. All the machine learning and deep learning algorithm experiments were run on Windows 10 with an Intel® Core™ i5-10400F CPU @ 2.90 GHz 12 CPUs and 512 GB of memory. The data analysis was conducted using Python 3.9.12 and IBM SPSS Statistics 22.

3. Results

3.1. Baseline information

The workflow of patient inclusion is displayed in Fig. 2. A total of 65,007 patients with RA were identified. Among them, 56,701 patients were aged ≥ 18 years. Of these, 46,138 patients had received RA drug treatments, and 4017 of these patients had at least 12 weeks of follow-up records. Patients with other rheumatic diseases (34 patients), pregnant or lactating women (9 patients), patients not using ZF (1964 patients), or patients lacking major research data (97 patients) were excluded. Ultimately, a total of 1913 patients were included in this study. Among them, there were 916 patients, 692 patients and 305 patients collected from the First Affiliated Hospital of Soochow University, Southern

Hospital of Southern Medical University and Hebei General Hospital, respectively.

According to a ratio of 4:1, 1530 patients were included in the training set and 383 patients were included in the testing set. The demographic and clinical characteristics of the patients in the dataset are displayed in Table 2. There were 1371 (71.67%) males and 542 (28.33%) females. The median age of the individuals in the whole dataset was 58 years (interquartile range [IQR] 50–68 years). Among the gender_age groups, middle-aged males had the highest proportion (37.11%), and young females had the lowest proportion (4.81%). The daily doses of ZF were divided into four subgroups: 60 mg (*n* [%] = 108 [5.65%]), 120 mg (*n* [%] = 900 [47.05%], 180 mg (*n* [%] = 192 [10.04%]), and 240 mg (*n* [%] = 713 [37.27%]). There were 599 (31.31%) patients who smoked and 371 (19.39%) who had CVD. For treatment plans, the number (%) treated with MTX + ZF, IGU + ZF, LEF + MTX + ZF, LEF + ZF, MTX + NSAIDs + ZF, TNF-α inhibitor + ZF, and JAK inhibitor + ZF was 468 (24.46%), 253 (13.23%), 263 (13.75%), 217 (11.34%), 220 (11.50%), 254 (13.28%), and 238 (12.44%), respectively.

There were 176 patients with abnormal liver function, their demographic and clinical characteristics are displayed in Table S2. Among them, 126 were males and 50 were females. The median age was 68.5 years (IQR 56–75 years). Among the gender_age groups, elderly males had the highest proportion (56.25%). In the daily dose subgroups, the dose of 120 mg was most commonly used (51.70%). About 49% of the patients smoked and 12.50% had CVD. MTX + ZF was the most commonly used treatment plan (31.25%).

3.2. Variable selection

The results of univariate analysis are shown in Table 3. A total of 22 significant variables were selected with *P* < 0.05, including age, gender_age, CVD status, treatment plan, ZF daily dose, CRP, RF, ESR, hemoglobin, prealbumin, uric acid, total protein, albumin, γ-glutamyl transferase, neutrophil absolute value, mean platelet volume, mean corpuscular volume, mean corpuscular hemoglobin concentration, mean corpuscular hemoglobin, packed cell volume, TC and TG.

After that, the 22 variables were further screened through the SFS method. Based on the XGBoost models developed with 1 to 22 variables, the F1_score of each XGBoost model was obtained (Fig. 3). As the number of included variables was increased, the F1_score of the model continued to increase and reached a maximum value when the model included 6 variables (F1_score = 0.73). As a concise and accurate model was pursued, a minimal variable set that can achieve good model performance was adopted. The first 6 variables were gender_age, treatment plan, age, TG, smoking status, and TC.

The F1_scores of the different models based on 22 and 6 variables are illustrated in Fig. 4 and Table S3. According to Fig. 4A, in the models based on 22 variables obtained by univariate analysis, the LightGBM model shows the best result with a mean F1_score of 0.77 and a standard deviation (Std) of 0.07. In Fig. 4B, for the models based on the 6 variables obtained by SFS, the LightGBM model yields the best results, with a mean F1_score of 0.78 and an Std of 0.07. A comparison of the models developed with 6 variables and 22 variables revealed that the model developed with 6 variables had a higher F1_score, which was both concise and robust. Therefore, the 6 variables were used for modeling in the testing set. The difference of the target variable and important influencing variables between training set and testing set is illustrated in Table S4. It can be seen that variables between the two datasets did not show a significant difference (*P* value > 0.05).

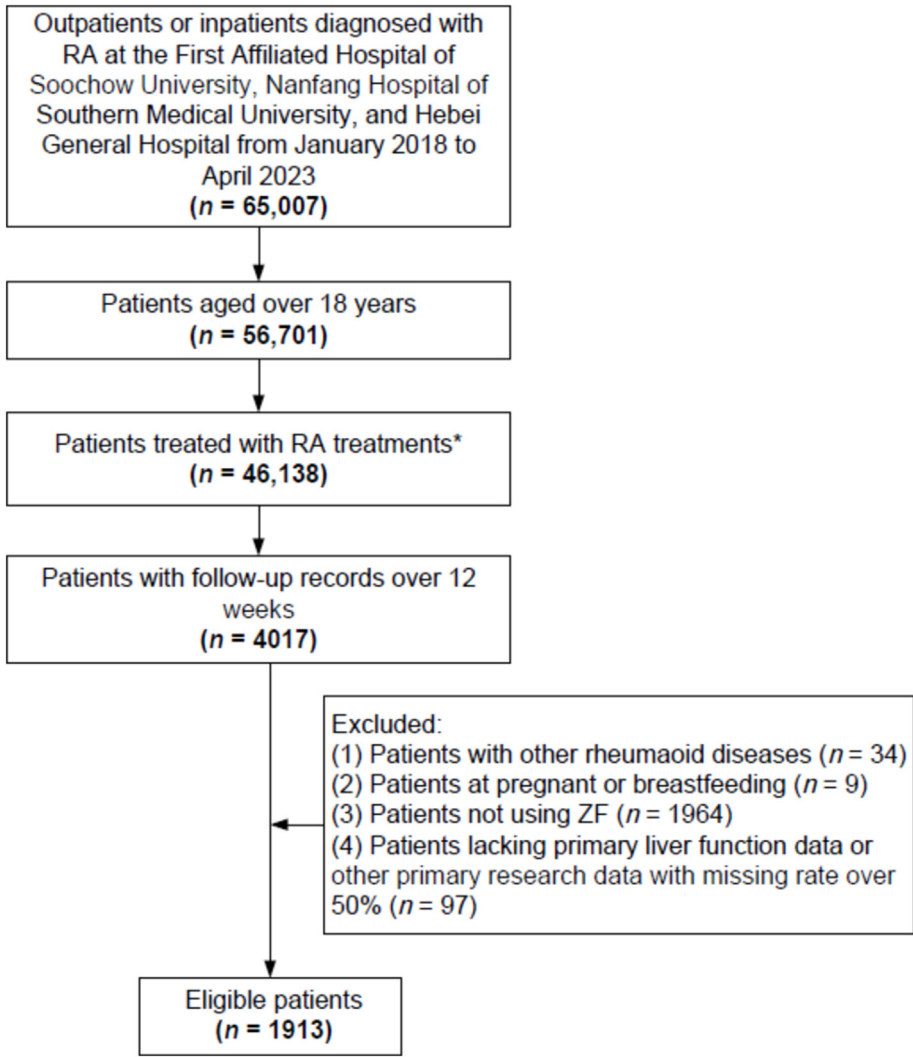


图2. 患者招募工作流程。RA：类风湿性关节炎；ZF：正清风痛宁。

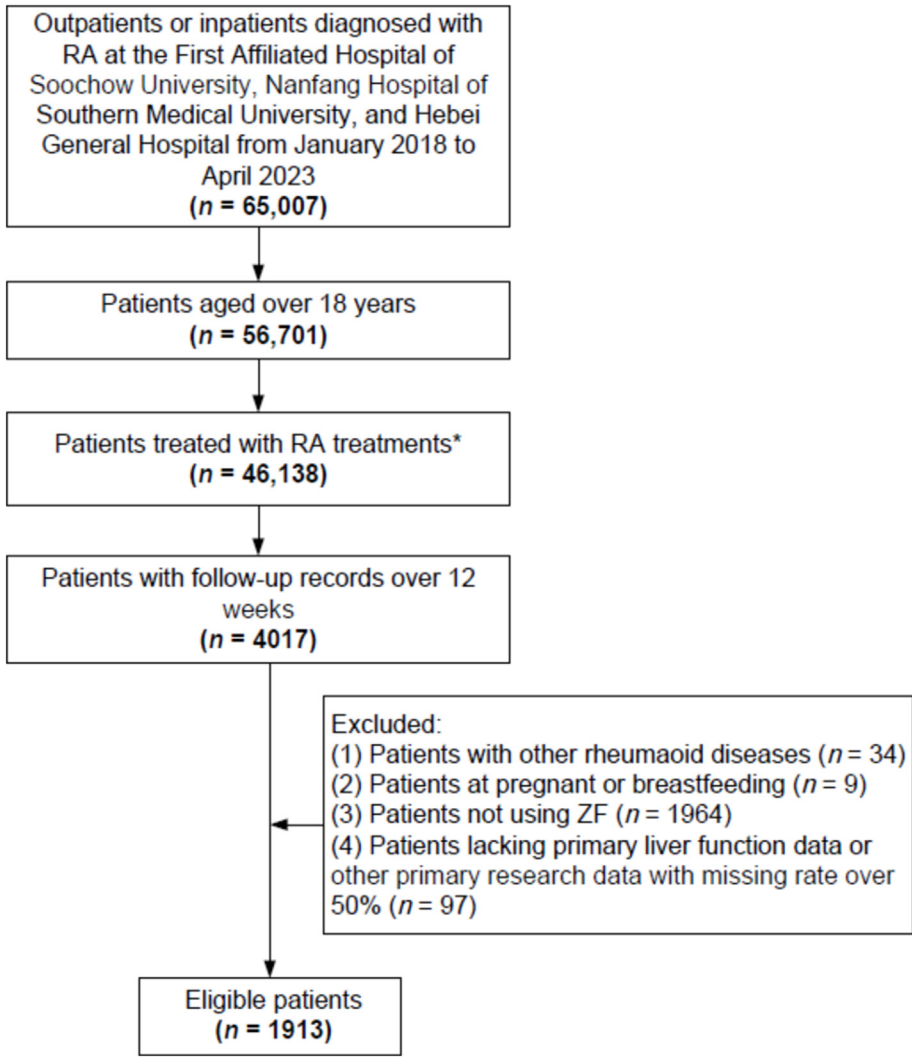


Fig. 2. Workflow of patient enrollment. RA: rheumatoid arthritis; ZF: Zhengqing Fengtongning.

3.3. 模型性能与解释

测试集中10个模型的预测性能如表4所示。其中，LightGBM模型表现最佳。该模型具有以下预测指标：预测ALT < 40 U/L时的精确度为 = 0.99，召回率为 = 0.97，F1_值为 = 0.98，AUC为 = 0.98，敏感性为 = 0.97，特异性为 = 0.85；预测ALT 40 < 80 U/L时的精确度为 = 0.60，召回率为 = 0.83，F1_值为 = 0.70，AUC为 = 0.98，敏感性为 = 0.83，特异性为 = 0.97；预测ALT 80 U/L时的精确度为 = 0.83，召回率为 = 0.63，F1_值为 = 0.71，AUC为 = 0.97，敏感性为 = 0.63，特异性为 = 1.00。整个LightGBM模型的准确率为0.96。

基于LightGBM模型，用于建模的6个变量的重要性评分如图S1和表S5所示。TC、TG、治疗方案、年龄、性别_年龄和吸烟状况的重要性评分分别为388、292、232、202、155和122。重要性评分表示特征在模型中的重要程度。评分越高，说明该变量对ZF导致的异常肝功能的预测能力越强。结果显示，TC是预测ZF导致的异常肝功能最重要的变量，其次是TG、治疗方案、年龄、性别_年龄和吸烟状况。22个变量的重要性评分显示在表S6中。

对于这6个重要变量，每个变量对模型输出的影响通过SHAP分析展示（图S2–S7）。SHAP散点图显示了变量与异常肝功能发生之间的单调关系。具体而言，随着TC和TG水平升高以及年龄增大，异常肝功能的危险增加。对于不同的治疗方案，当使用TNF-α抑制剂 + ZF、JAK抑制剂 + ZF、MTX + NSAIDs + ZF或LEF + ZF时，异常肝功能的危险最容易增加。对于年龄_性别群体，异常肝功能的危险在老年患者中最大，其次是中年女性。此外，吸烟患者的异常肝功能风险高于非吸烟患者。

测试集由383名患者组成，其中357名患者ALT为 < 40 U/L，18名患者ALT为40 < 80 U/L，8名患者ALT为80 U/L（图S8）。基于最终的LightGBM模型，预测了异常肝功能的危险，并计算了精确率和召回率。精确率是模型正确预测特定结果的能力，而召回率是模型对所有阳性病例进行正确预测的能力，即不遗漏阳性病例。352名患者被预测属于ALT < 40 U/L组。其中348名患者分类正确，4名患者分类错误，9名患者分类错误，从而得到99%的精确率和

3.3. Model performance and interpretation

The predictive performance of the 10 models in the testing set is illustrated in Table 4. Among them, the LightGBM model shows the best performance. The model has the following predictive metrics: precision = 0.99, recall rate = 0.97, F1_score = 0.98, AUC = 0.98, sensitivity = 0.97 and specificity = 0.85 for predicting ALT < 40 U/L; precision = 0.60, recall rate = 0.83, F1_score = 0.70, AUC = 0.98, sensitivity = 0.83 and specificity = 0.97 for predicting 40 ≤ ALT < 80 U/L; and precision = 0.83, recall rate = 0.63, F1_score = 0.71, AUC = 0.97, sensitivity = 0.63 and specificity = 1.00 for predicting ALT ≥ 80 U/L. The accuracy is 0.96 for the whole LightGBM model.

Based on the LightGBM model, the importance scores of the 6 variables used for modeling are presented in Fig. S1 and Table S5. The importance scores of TC, TG, treatment plan, age, gender_age and smoking status were 388, 292, 232, 202, 155 and 122, respectively. The importance score indicates the degree of importance of features in the model. A higher score means greater predictive power of the variable on the ZF-induced abnormal liver function. The results reveal that TC was the most predictive variable of ZF-induced abnormal liver function, followed by TG, treatment plan, age, gender_age and smoking status. The importance scores of the 22 variables are displayed in Table S6.

For the 6 important variables, each variable's impact on the model output was demonstrated via SHAP analysis (Figs. S2–S7). The SHAP scatter plots indicate monotonic relationships between variables and the occurrence of abnormal liver function. Specifically, the risk of abnormal liver function increases with increasing TC and TG levels, and greater age. For diverse treatment plans, the risk of abnormal liver function was most likely to increase when TNF-α inhibitor + ZF, JAK inhibitor + ZF, MTX + NSAIDs + ZF, or LEF + ZF was used. For the age_gender groups, the risk of abnormal liver function was greatest in elderly patients, followed by middle-aged females. Additionally, the risk of abnormal liver function was greater in smoking patients than non-smoking ones.

The testing set consisted of 383 patients, among which 357 had ALT < 40 U/L, 18 had 40 ≤ ALT < 80 U/L, and 8 had ALT ≥ 80 U/L (Fig. S8). Based on the final LightGBM model, the risk of abnormal liver function was predicted, and the precision and recall rate were calculated. Precision is the ability of the model to correctly predict specific outcomes, while recall rate is the ability of the model to make correct predictions for all positive cases, that is, not to miss positive cases. Three hundred and fifty-two (352) patients were predicted to belong to the ALT < 40 U/L group. Of these 348 were correctly classified, 4 patients were incorrectly classified, and 9 patients were mis-classified, resulting in a precision of 99% and a

表 2 全数据集的基线信息。

类别	变量	中位数 (四分位距) 或 n [%]	未命中率
人口统计信息	年龄 (年)	58.00 (50.00–68.00)	0.00%
	性别		0.00%
	Male	1371 (71.67%)	
	女性	542 (28.33%)	
	性别_年龄		0.00%
	年轻男性	202 (10.56%)	
	年轻女性	92 (4.81%)	
	中年男子	710 (37.11%)	
	中年女性	260 (13.59%)	
	年长男性	459 (23.99%)	
ZF信息	年长女性	190 (9.93%)	
	每日剂量的ZF		0.00%
	60 毫克	108 (5.65%)	
	120 毫克	900 (47.05%)	
	180 毫克	192 (10.04%)	
吸烟史	240 毫克	713 (37.27%)	
	吸烟	599 (31.31%)	0.00%
心血管疾病史	CVD	371 (19.39%)	0.00%
药物信息	治疗方案		0.00%
	MTX + ZF	468 (24.46%)	
	IGU +ZF	253 (13.23%)	
	LEF + MTX + ZF	26 (13.75%)	
	LEF +ZF	217 (11.34%)	
	ZF +MTX + 非甾体抗炎药	220 (11.50%)	
	TNF-α抑制剂 +ZF	254 (13.28%)	
	JAK抑制剂 + ZF	238 (12.44%)	
	C反应蛋白 (毫克/升)	3.16 (0.83–8.93)	0.00%
	类风湿因子 (IU/mL)	100.31 (28.55–199.32)	0.00%
化验指标	红细胞沉降率 (毫米/小时)	25.00 (15.61–41.34)	0.00%
	球蛋白 (克/升)	28.10 (25.60–30.60)	22.74%
	血红蛋白 (克/升)	127.00 (119.00–135.75)	6.85%
	PA (毫克/升)	244.00 (215.00–276.00)	0.63%
	尿酸 (μmol/L)	287.55 (230.70–345.50)	36.64%
	总蛋白 (克/升)	71.20 (68.10–74.00)	0.58%
	白蛋白 (克/升)	42.90 (40.80–44.90)	22.74%
	γ-GT (U/L)	16.40 (14.20–24.40)	0.58%
	中性粒细胞 (10 ⁹ /L)	3.90 (2.92–5.09)	6.95%
	单核细胞 (10 ⁹ /L)	0.40 (0.31–0.51)	6.95%
	BC (10 ⁹ /L)	0.03 (0.02–0.04)	6.95%
	电导率 (10 ⁹ /升)	0.09 (0.06–0.16)	6.95%
	平均血小板体积 (fL)	10.00 (9.20–10.80)	7.27%
	LF (10 ⁹ /L)	1.54 (1.20–1.95)	6.95%
	红细胞分布宽度 (%)	13.20 (12.70–14.20)	6.85%
	平均红细胞体积 (fL)	91.20 (88.10–94.40)	6.85%
	平均红细胞血红蛋白浓度 (克/升)	327.00 (320.00–334.00)	6.85%
	平均红细胞血红蛋白量 (pg)	29.90 (28.70–31.30)	6.85%
	HCT (升/升)	0.39 (0.36–0.41)	6.85%
	红细胞计数 (10 ¹² /L)	4.27 (3.99–4.58)	6.85%
	血小板分布宽度 (%)	15.70 (11.70–16.30)	7.27%
	PCT (%)	0.22 (0.19–0.26)	7.42%
	血小板计数 (10 ⁹ /L)	222.00 (187.00–270.25)	6.85%
	LDL-C (毫摩尔/升)	2.22 (1.71–2.71)	62.00%
	总胆固醇 (毫摩尔/升)	3.14 (1.70–4.41)	0.00%
	甘油三酯 (mmol/L)	1.1 (0.87–1.52)	15.17%
	Cr (μ摩尔/升)	60.10 (51.70–76.30)	48.41%
	HDLC (毫摩尔/升)	0.65 (0.37–0.94)	4.29%
	DAS28-CRP	4.22 (3.69–5.43)	0.00%

ALB: 白蛋白; BC: 嗜碱性粒细胞计数; Cr: 肌酐; CRP: C反应蛋白; CVD: 心血管疾病; DAS28-CRP: 基于CRP水平的28关节疾病活动评分; EC: 嗜酸性粒细胞计数; ESR: 红细胞沉降率; γ-GT: γ-谷氨酰转移酶; GLB: 球蛋白; HB: 血红蛋白; HCT: 血细胞比容; HDLC: 高密度脂蛋白胆固醇; IGU: 伊古替莫德; IQR: 四分位数范围; JAK: JAK激酶; LDLC: 低密度脂蛋白胆固醇; LEF: 来氟米特; LF: 淋巴细胞计数; MCH: 平均红细胞血红蛋白量; MCHC: 平均红细胞血红蛋白浓度; MCV: 平均红细胞体积; MONO: 单核细胞计数; MPV: 平均血小板体积; MTX: 甲氨蝶呤; NEUT: 中性粒细胞计数; NSAID: 非甾体抗炎药; PA: 前白蛋白; PCT: 血小板压积; PDW: 血小板分布宽度; PLT: 血小板计数; RBC: 红细胞计数; RDW: 红细胞分布宽度; RF: 类风湿因子; TC: 总胆固醇; TG: 甘油三酯; TNF: 肿瘤坏死因子; TP: 总蛋白

召回率为97%。预测有25名患者属于40 ALT < 80 U/L组。其中，15名被正确分类，10名分类错误，3名患者被误分类；精确率和召回率分别为60%和83%。有6名患者被正确预测为ALT 80 U/L组，1名患者分类错误，3名患者被误分类；精确率和召回率分别为83%和63%。精确率和召回率的结果表明

异常肝功能的发生被准确预测，模型效果良好，能够满足临床需求。

3.4. 外部验证

外部验证集包含106例患者。其中，99例患者的ALT为 < 40 U/L，5例患者的ALT为40 < 80 U/L，2例患者的ALT为80 U/L（图S9）。

Table 2
Baseline information for the whole dataset.

Category	Variable	Median (IQR) or n [%]	Miss rate
Demographic information	Age (year)	58.00 (50.00–68.00)	0.00%
	Gender		0.00%
	Male	1371 (71.67%)	
	Female	542 (28.33%)	
	Gender_age		0.00%
	Young male	202 (10.56%)	
	Young female	92 (4.81%)	
	Middle-aged male	710 (37.11%)	
	Middle-aged female	260 (13.59%)	
	Elderly male	459 (23.99%)	
ZF information	Elderly female	190 (9.93%)	
	Daily dose of ZF		0.00%
	60 mg	108 (5.65%)	
	120 mg	900 (47.05%)	
	180 mg	192 (10.04%)	
Smoking history	240 mg	713 (37.27%)	
	Smoking	599 (31.31%)	0.00%
CVD history	CVD	371 (19.39%)	0.00%
Medication information	Treatment plan		0.00%
	MTX + ZF	468 (24.46%)	
	IGU + ZF	253 (13.23%)	
	LEF + MTX + ZF	26 (13.75%)	
	LEF + ZF	217 (11.34%)	
	ZF + MTX + NSAIDs	220 (11.50%)	
	TNF-α inhibitor + ZF	254 (13.28%)	
	JAK inhibitor + ZF	238 (12.44%)	
	CRP (mg/L)	3.16 (0.83–8.93)	0.00%
	RF (IU/mL)	100.31 (28.55–199.32)	0.00%
Assay index	ESR (mm/h)	25.00 (15.61–41.34)	0.00%
	GLB (g/L)	28.10 (25.60–30.60)	22.74%
	HB (g/L)	127.00 (119.00–135.75)	6.85%
	PA (mg/L)	244.00 (215.00–276.00)	0.63%
	UA (μmol/L)	287.55 (230.70–345.50)	36.64%
	TP (g/L)	71.20 (68.10–74.00)	0.58%
	ALB (g/L)	42.90 (40.80–44.90)	22.74%
	γ-GT (U/L)	16.40 (14.20–24.40)	0.58%
	NEUT (10 ⁹ /L)	3.90 (2.92–5.09)	6.95%
	MONO (10 ⁹ /L)	0.40 (0.31–0.51)	6.95%
	BC (10 ⁹ /L)	0.03 (0.02–0.04)	6.95%
	EC (10 ⁹ /L)	0.09 (0.06–0.16)	6.95%
	MPV (fL)	10.00 (9.20–10.80)	7.27%
	LF (10 ⁹ /L)	1.54 (1.20–1.95)	6.95%
	RDW (%)	13.20 (12.70–14.20)	6.85%
	MCV (fL)	91.20 (88.10–94.40)	6.85%
	MCHC (g/L)	327.00 (320.00–334.00)	6.85%
	MCH (pg)	29.90 (28.70–31.30)	6.85%
	HCT (L/L)	0.39 (0.36–0.41)	6.85%
	RBC (10 ¹² /L)	4.27 (3.99–4.58)	6.85%
Other examinations	PDW (%)	15.70 (11.70–16.30)	7.27%
	PCT (%)	0.22 (0.19–0.26)	7.42%
	PLT (10 ⁹ /L)	222.00 (187.00–270.25)	6.85%
	LDLC (mmol/L)	2.22 (1.71–2.71)	62.00%
	TC (mmol/L)	3.14 (1.70–4.41)	0.00%
	TG (mmol/L)	1.1 (0.87–1.52)	15.17%
	Cr (μmol/L)	60.10 (51.70–76.30)	48.41%
	HDLC (mmol/L)	0.65 (0.37–0.94)	4.29%
	DAS28-CRP	4.22 (3.69–5.43)	0.00%

ALB: albumin; BC: basophil count; Cr: creatinine; CRP: C-reactive protein; CVD: cardiovascular disease; DAS28-CRP: disease activity score in 28 joints based on CRP level; EC: eosinophil count; ESR: erythrocyte sedimentation rate; γ-GT: γ-glutamyl transferase; GLB: globulin; HB: hemoglobin; HCT: hematocrit; HDLC: high-density lipoprotein cholesterol; IGU: iguratimod; IQR: interquartile range; JAK: Janus kinase; LDLC: low-density lipoprotein cholesterol; LEF: leoflunomide; LF: lymphocyte count; MCH: mean corpuscular hemoglobin; MCHC: mean corpuscular hemoglobin concentration; MCV: mean corpuscular volume; MONO: monocyte count; MPV: mean platelet volume; MTX: methotrexate; NEUT: neutrophil count; NSAID: nonsteroidal anti-inflammatory drug; PA: prealbumin; PCT: plateletocrit; PDW: platelet distribution width; PLT: platelet count; RBC: red blood cell count; RDW: red blood cell distribution width; RF: rheumatoid factor; TC: total cholesterol; TG: triglyceride; TNF: tumor necrosis factor; TP: total protein; UA: uric acid; ZF: Zhengqing Fengtongning.

recall rate of 97%. Twenty-five patients were predicted to be in the 40 ≤ ALT < 80 U/L group. Of these, 15 were correctly classified, 10 were incorrectly classified, and 3 patients were mis-classified; the precision and recall rate were 60% and 83%, respectively. Six patients were correctly predicted to be in the ALT ≥ 80 U/L group, 1 patient was incorrectly classified, and 3 patients were mis-classified; the precision and recall rate were 83% and 63%, respectively. The results of precision and recall rate indicate that the

occurrence of abnormal liver function was predicted accurately and the model had a good effect to meet the clinical needs.

3.4. External validation

The external validation set contained 106 patient cases. Among them, there were 99 patients with ALT < 40 U/L, 5 patients with 40 ≤ ALT < 80 U/L, and 2 patients with ALT ≥ 80 U/L (Fig. S9).

表 3 单变量分析。

变量	统计	P 值	变量	统计	P 值
Age	62.649	< 0.001	BC	4.599	0.100
性别	0.240	0.886	MONO	2.140	0.343
性别_年龄	145.348	< 0.001	EC	4.151	0.125
吸烟	33.161	< 0.001	MPV	6.906	0.032
CVD	6.549	0.0378	LF	4.051	0.132
治疗方案	150.466	< 0.001	RDW	4.276	0.118
每日剂量的ZF	11.305	0.079	MCV	7.558	0.023
CRP	7.008	0.030	MCHC	16.983	< 0.001
RF	7.352	0.025	MCH	16.699	< 0.001
ESR	8.657	0.013	PCV	16.895	< 0.001
DAS28-CRP	0.936	0.626	RBC	3.844	0.146
GLB	2.138	0.343	PDW	1.585	0.453
HB	19.355	< 0.001	PCT	4.514	0.105
PA	8.882	0.012	PLT	3.083	0.214
UA	6.909	0.032	LDLC	2.871	0.238
TP	8.591	0.014	TC	220.098	< 0.001
ALB	10.409	0.005	TG	14.704	0.001
γ-GT	74.655	< 0.001	Cr	0.464	0.793
NEUT	10.482	0.005	HDLC	0.724	0.696

ALB: 白蛋白; BC: 嗜碱细胞计数; Cr: 肌酐; CRP: C 反应蛋白; CVD: 心血管疾病; DAS28-CRP: 基于 CRP 水平的 28 关节疾病活动评分; EC: 嗜酸细胞计数; ESR: 红细胞沉降率; c-GT: γ-谷氨酰转移酶; GLB: 球蛋白; HB: 血红蛋白; HCT: 红细胞比容; HDLC: 高密度脂蛋白胆固醇; IGU: 伊古利莫德; JAK: JAK 激酶; LDLC: 低密度脂蛋白胆固醇; LEF: 来氟米特; LF: 淋巴细胞计数; MCH: 平均红细胞血红蛋白量; MCHC: 平均红细胞血红蛋白浓度; MCV: 平均红细胞体积; MONO: 单核细胞计数; MPV: 平均血小板体积; MTX: 甲氨蝶呤; NEUT: 中性粒细胞计数; NSAID: 非甾体抗炎药; PA: 前白蛋白; PCT: 血小板压积; PDW: 血小板分布宽度; PLT: 血小板计数; RBC: 红细胞计数; RDW: 红细胞分布宽度; RF: 类风湿因子; TC: 总胆固醇; TG: 甘油三酯; TNF: 肿瘤坏死因子; TP: 总蛋白; UA: 尿酸

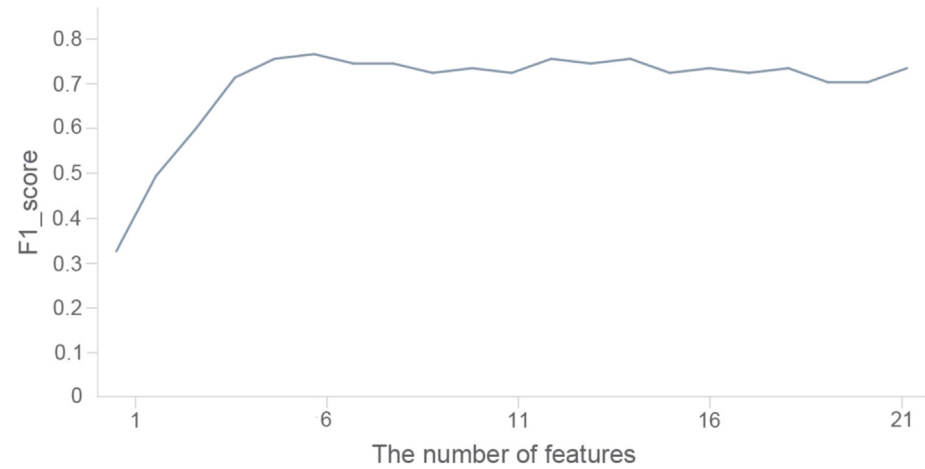


图 3. 与顺序前向选择中排序特征数量对应的 F1_评分。

我们使用该模型预测了肝功能水平，并从混淆矩阵中计算了模型的汇总统计数据。共有101名患者被预测属于ALT < 40 U/L组。其中99名被正确分类，2名分类错误，没有患者被误分类。该模型在该组的精确度为98%，召回率为100%。有3名患者被正确预测属于40 ALT < 80 U/L组，2名患者被误分类，导致该组的精确度为100%，召回率为60%。有2名患者被正确预测属于ALT 80 U/L组，没有患者被误分类至该组，因此该组的精确度为100%，召回率为100%。精确度和召回率的结果表明，在验证数据集中异常肝功能的发生被准确且全面地预测出来。

3.5. 模型应用

我们建立了一个基于LightGBM模型的ZF诱导异常肝功能预测的网络应用（图5）。

通过使用该网络应用程序，用户可以在网页上输入六个重要变量的数值，从而预测ZF引起的异常肝功能的风险。在将模型应用于数据后，结果页面将显示提交数据的预测肝功能值。

4. 讨论

类风湿性关节炎（RA）是最常见的免疫性疾病之一；目前尚无法治愈，需要终身治疗。抗风湿药物在临床上常用于缓解症状和改善生活质量，但长期使用容易引起不良反应（ADR）。中药制剂ZF于2000年在中国上市，并广泛用于RA的治疗。随着中医药在临床上的广泛应用，人们开始关注ZF相关的不良反应。ZF说明书中未将肝功能异常列为潜在不良反应，但对包含956名受试者的11项研究进行回顾发现，肝功能异常是最常见的不良反应之一。

Table 3
Univariate analysis.

Variable	Statistics	P value	Variable	Statistics	P value
Age	62.649	< 0.001	BC	4.599	0.100
Gender	0.240	0.886	MONO	2.140	0.343
Gender_age	145.348	< 0.001	EC	4.151	0.125
Smoking	33.161	< 0.001	MPV	6.906	0.032
CVD	6.549	0.0378	LF	4.051	0.132
Treatment plan	150.466	< 0.001	RDW	4.276	0.118
Daily dose of ZF	11.305	0.079	MCV	7.558	0.023
CRP	7.008	0.030	MCHC	16.983	< 0.001
RF	7.352	0.025	MCH	16.699	< 0.001
ESR	8.657	0.013	PCV	16.895	< 0.001
DAS28-CRP	0.936	0.626	RBC	3.844	0.146
GLB	2.138	0.343	PDW	1.585	0.453
HB	19.355	< 0.001	PCT	4.514	0.105
PA	8.882	0.012	PLT	3.083	0.214
UA	6.909	0.032	LDLC	2.871	0.238
TP	8.591	0.014	TC	220.098	< 0.001
ALB	10.409	0.005	TG	14.704	0.001
γ-GT	74.655	< 0.001	Cr	0.464	0.793
NEUT	10.482	0.005	HDLC	0.724	0.696

ALB: albumin; BC: basophil count; Cr: creatinine; CRP: C-reactive protein; CVD: cardiovascular disease; DAS28-CRP: disease activity score in 28 joints based on CRP level; EC: eosinophil count; ESR: erythrocyte sedimentation rate; γ-GT: γ-glutamyl transferase; GLB: globulin; HB: hemoglobin; HCT: hematocrit; HDLC: high-density lipoprotein cholesterol; IGU: iguratimod; JAK: Janus kinase; LDLC: low-density lipoprotein cholesterol; LEF: leoflunomide; LF: lymphocyte count; MCH: mean corpuscular hemoglobin; MCHC: mean corpuscular hemoglobin concentration; MCV: mean corpuscular volume; MONO: monocyte count; MPV: mean platelet volume; MTX: methotrexate; NEUT: neutrophil count; NSAID: nonsteroidal anti-inflammatory drug; PA: prealbumin; PCT: plateletocrit; PDW: platelet distribution width; PLT: platelet count; RBC: red blood cell count; RDW: red blood cell distribution width; RF: rheumatoid factor; TC: total cholesterol; TG: triglyceride; TNF: tumor necrosis factor; TP: total protein; UA: uric acid; ZF: Zhengqing Fengtongning.

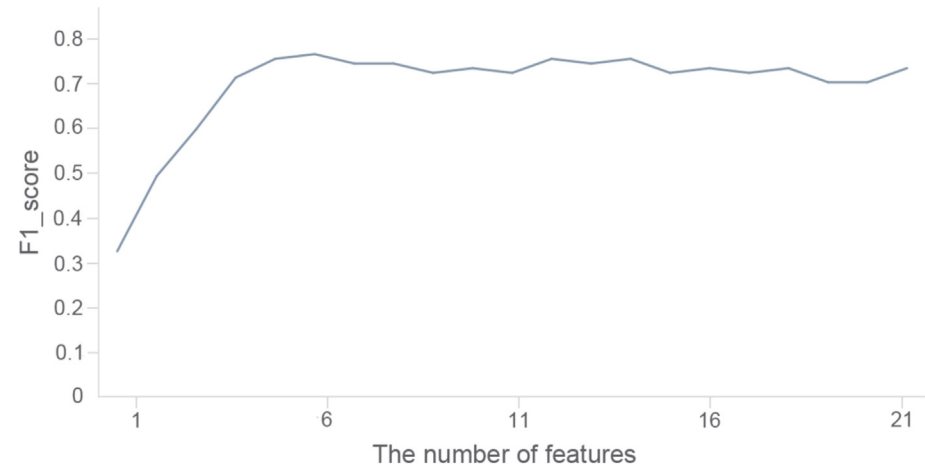


Fig. 3. The F1_score corresponding to the number of ranked features in sequential forward selection.

We predicted the levels of liver function using the model and calculated summary statistics for the model from the confusion matrix. One hundred and one (101) patients were predicted to belong to the ALT < 40 U/L group. Ninety-nine were correctly classified, 2 were incorrectly classified, and no patients were mis-classified. The model had a precision of 98% and a recall rate of 100% for this group. Three patients were correctly predicted to belong to the $40 \leq \text{ALT} < 80$ U/L group, and 2 patients were mis-classified, resulting in a precision of 100% and a recall rate of 60%. Two patients were correctly predicted to belong to the $\text{ALT} \geq 80$ U/L group, and no patients were mis-classified to this group, resulting in a precision of 100% and a recall rate of 100%. The results of precision and recall rate indicate that the occurrence of abnormal liver function was predicted accurately and comprehensively in the validation dataset.

3.5. Model application

We established a web application for predicting ZF-induced abnormal liver function based on the LightGBM model (Fig. 5).

By using the web application, users can predict the risk of ZF-induced abnormal liver function by entering the values of the six important variables on the web page. After the model is applied to the data, the result page will display the predicted liver function value for the submitted data.

4. Discussion

RA is one of the most common immune diseases; currently, it cannot be cured and requires lifelong treatment [26]. Anti-rheumatic drugs are often used in clinical practice to relieve symptoms and improve quality of life, but their long-term use can easily cause ADRs [27,28]. ZF was launched in China in 2000 and has been widely used in the treatment of RA [29]. With the widespread clinical use of traditional Chinese medicine, attention has been given to the ADRs associated with ZF [30–34]. Abnormal liver function was not mentioned in the instructions for ZF as a potential ADR, but a review of 11 studies containing 956 participants revealed that abnormal liver function is one of the most common ADRs

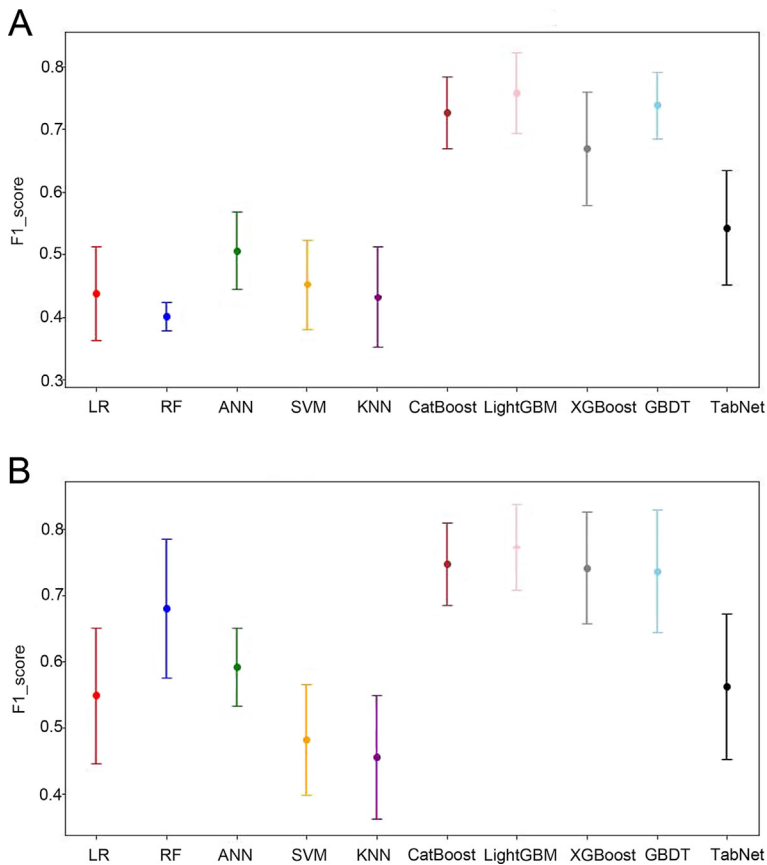


图4. 基于22个变量(A)和6个变量(B)的不同模型的F1分数, 附平均值 $\pm$ 标准差。ANN: 人工神经网络; CatBoost: 类别提升; GBDT: 梯度提升决策树; KNN: k近邻; LightGBM: 轻量级梯度提升机; LR: 逻辑回归; RF: 随机森林; SVM: 支持向量机; XGBoost: 极端梯度提升。

与ZF [29,35]相关。肝功能异常可能导致严重的临床结果, 特别是在RA患者的长期治疗中。因此, 有必要利用ZF预测RA患者肝功能异常的风险。识别影响ZF不良反应的因素并采取早期预防措施以确保药物安全使用至关重要。然而, 目前基于真实世界证据(RWE)的ZF安全性研究仍然缺乏。本研究关注使用ZF治疗的RA患者, 并从其临床记录中收集了RWE。我们确定了几个对肝功能异常风险有重要影响的变量, 并使用机器学习技术开发了预测模型。

在本研究中, 多种变量显示对异常肝功能的风险具有显著影响。根据SHAP分析结果, 随着总胆固醇(TC)和甘油三酯(TG)水平的升高, 异常肝功能的风险增加。TC和TG水平与肝功能密切相关[36]。肝脏是代谢TC和TG的主要器官。如果肝功能异常, TC和TG的合成和清除可能会受到影响, 从而导致血液中TC和TG水平升高[36]。因此, 在使用ZF治疗时, 总胆固醇和甘油三酯水平升高的患者更容易发生异常肝功能; 这一知识可促使根据预测结果及时调整治疗方案, 例如减少ZF的剂量。此外, TNF- α 抑制剂 + ZF、JAK抑制剂 + ZF、甲氨蝶呤(MTX) + 非甾体抗炎药(NSAIDs) + ZF以及来氟米特(LEF) + ZF治疗方案增加了异常肝功能的风险。因为具有免疫调节和抗炎作用

从而影响药物的有效性和安全性, 并导致不良反应(ADR) [37,38]。在临床环境中, 使用TNF- α 抑制剂 + ZF、JAK抑制剂 + ZF、MTX + 、NSAIDs + ZF和LEF + ZF治疗方案的患者需要注意其发生肝功能异常的风险。由于条件有限, 我们未进一步研究药物联合使用对ZF引起肝功能异常影响的潜在原因, 这可能与药物-药物相互作用有关, 应在未来研究中深入探讨。此外, 肝功能异常的风险在吸烟患者中最可能增加。吸烟会增加促炎细胞因子(白细胞介素 [IL-1], IL-6和TNF- α)的产生, 这些因子会导致肝细胞损伤, 并可能降低肝功能[39]。因此, 吸烟患者在接受ZF治疗期间需要特别注意肝功能异常。Wu等人[40] 基于4,064名患者发现, 与相关ADR的发生有关

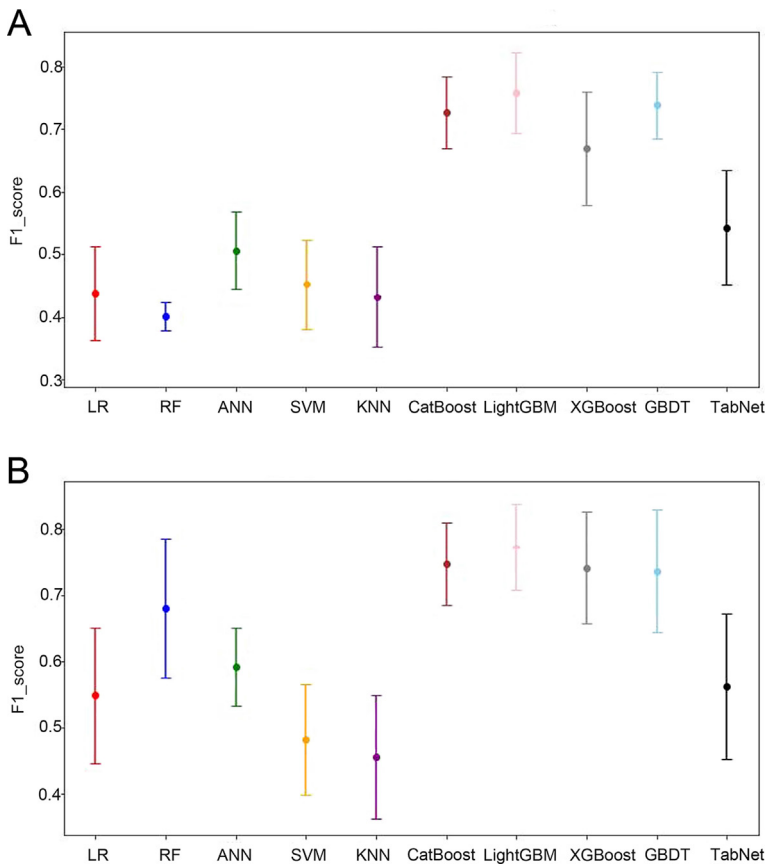


Fig. 4. F1-score of different models based on 22 variables (A) and 6 variables (B) with the mean $\pm$ standard deviation. ANN: artificial neural network; CatBoost: categorical boosting; GBDT: gradient boosting decision tree; KNN: k-nearest neighbor; LightGBM: light gradient boosting machine; LR: logistic regression; RF: random forest; SVM: support vector machine; XGBoost: extreme gradient boosting.

associated with ZF [29,35]. Abnormal liver function may lead to severe clinical outcomes, especially in long-term treatment in RA patients. Therefore, it is necessary to predict the risk of abnormal liver function in RA patients using ZF. It is crucial to recognize the factors influencing the ADRs of ZF and to adopt early prevention measures for safe drug use. However, at present, studies based on RWE on the safety of ZF are lacking. This study focused on RA patients treated with ZF and collected RWE from their clinical records. We identified several important variables that contribute to the risk of abnormal liver function and used a machine learning technique to develop a predictive model.

In this study, multiple variables were shown to have remarkable impacts on the risk of abnormal liver function. According to the results of the SHAP analysis, the risk of abnormal liver function increased with increasing TC and TG levels. TC and TG levels are strongly associated with liver function [36]. The liver is the main organ that metabolizes TC and TG. If the function of the liver is abnormal, the synthesis and clearance of TC and TG may be affected, resulting in an increase in blood level of TC and TG [36]. Thus, abnormal liver function may be more likely to occur in patients with elevated TC and TG levels when using ZF treatment; this knowledge may facilitate timely adjustments to treatment regimen based on the prediction results, such as reducing the dose of ZF. Furthermore, the TNF- α inhibitor + ZF, JAK inhibitor + ZF, MTX + NSAIDs + ZF, and LEF + ZF treatment regimens increased the risk of abnormal liver function. Because of the immunomodulatory and anti-inflammatory effects of ZF, it is commonly used in combination with other drugs for treating RA in the clinic. Drug interactions caused by the combination of drugs in treatment plans may either change the strength or the property of the drug effect,

thus affecting the effectiveness and safety of the drug and lead to ADRs [37,38]. In the clinical setting, patients using TNF- α inhibitor + ZF, JAK inhibitor + ZF, MTX + NSAIDs + ZF, and LEF + ZF treatment regimens need to be aware of their risk of developing abnormal liver function. Due to limited conditions, we did not further study the underlying reasons of the effect of drug combination on ZF-induced abnormal liver function, which may be related to drug-drug interaction and should be deeply explored in future study. Additionally, the risk of abnormal liver function was most likely to increase in smoking patients. Smoking increases the production of proinflammatory cytokines (interleukin [IL-1], IL-6 and TNF- α), which cause liver cell damage and may decrease liver function [39]. Thus, smoking patients need to take extra precautions against abnormal liver function during ZF treatment. Wu et al. [40] found that based on 4,064 patients, the occurrence of ADRs associated with ZF was related to gender, specific age group, and drug combination. The authors demonstrated that the risk of ADRs associated with ZF was greater in females and people aged 31 to 50 years [40]. In this study, we investigated the relationships between gender, age and ZF-induced abnormal liver function in diverse gender_age groups. Notably, abnormal liver function is more likely to occur in elderly individuals than in young individuals. Among middle-aged (45–65 years old) patients, the risk of abnormal liver function is greater in females than in males. This may be due to decreases in the physiological activity of various systems and reduced ability to metabolize drugs in the elderly [41]. Moreover, the physiological status of females is different from that of males; females are more sensitive to drugs, have lower tolerance, and typically take a more active role in managing their health [8]. Therefore, the model constructed

表4 模型性能。

模型	ALT 组	精确	召回率	F1 分数	AUC	敏感度	特异性	准确性
LR	0	0.95	0.99	0.97	0.91	0.99	0.35	0.94
	1	0.33	0.11	0.17	0.86	0.11	0.99	
	2	0.80	0.50	0.62	0.99	0.50	1.00	
RF	0	0.95	1.00	0.98	0.98	1.00	0.31	0.95
	1	0.57	0.22	0.32	0.97	0.22	0.99	
	2	1.00	0.13	0.22	0.99	0.13	1.00	
ANN	0	0.95	0.99	0.97	0.95	0.99	0.31	0.93
	1	0.14	0.06	0.08	0.92	0.06	0.98	
	2	0.75	0.38	0.50	0.99	0.38	1.00	
SVM	0	0.94	1.00	0.97	0.95	1.00	0.12	0.94
	1	0.00	0.00	0.00	0.92	0.00	1.00	
	2	0.75	0.38	0.50	0.99	0.38	1.00	
KNN	0	0.96	0.97	0.97	0.90	0.97	0.42	0.92
	1	0.25	0.28	0.26	0.81	0.28	0.96	
	2	0.00	0.00	0.00	0.87	0.00	1.00	
CatBoost	0	0.98	0.97	0.98	0.98	0.97	0.69	0.95
	1	0.55	0.61	0.58	0.97	0.61	0.98	
	2	0.86	0.75	0.80	0.99	0.75	1.00	
LightGBM	0	0.99	0.97	0.98	0.98	0.97	0.85	0.96
	1	0.60	0.83	0.70	0.98	0.83	0.97	
	2	0.83	0.63	0.71	0.97	0.63	1.00	
XGBoost	0	0.95	1.00	0.97	0.98	1.00	0.23	0.94
	1	0.67	0.22	0.33	0.97	0.22	0.99	
	2	0.00	0.00	0.00	0.99	0.00	1.00	
GBDT	0	0.99	0.96	0.97	0.98	0.96	0.81	0.95
	1	0.50	0.78	0.61	0.97	0.78	0.96	
	2	0.71	0.63	0.67	0.97	0.63	0.99	
TabNet	0	0.96	0.99	0.97	0.94	0.99	0.38	0.93
	1	0.27	0.17	0.21	0.92	0.17	0.98	
	2	0.75	0.38	0.50	0.92	0.38	1.00	

标签“0”表示ALT为 < 40 U/L 的患者，标签“1”表示ALT为 40 < 80 U/L 的患者，标签“2”表示ALT为 80 U/L 的患者。ALT: 丙氨酸转氨酶；ANN: 人工神经网络；AUC: 曲线下面积；CatBoost: 类别提升；GBDT: 梯度提升决策树；KNN: K近邻；LightGBM: 轻量梯度提升机；LR: 逻辑回归；RF: 随机森林；SVM: 支持向量机；XGBoost: 极端梯度提升。

本研究对老年患者和中年女性患者更有益。模型预测使患者在使用ZF帮助管理RA时，能够意识到其肝功能潜在的风险。

本研究比较了10种机器学习和深度学习算法。在比较预测指标后，选择LightGBM来建模异常肝功能的风险。LightGBM是GBDT框架的改进版本，采用基于决策树的快速、分布式和高性能方法[13,42]。LightGBM使用三种技术——基于梯度的单侧采样、特征独占捆绑和直方图分析，以减少所需样本数、特征数量和内存消耗[13,42]。LightGBM已针对并行计算进行了优化，支持特征并行和数据并行，并对每种并行模式进行了优化以减少通信开销[13,42]。在临床应用中，机器学习和深度学习技术可以用于开发具有较强泛化能力和良好稳健性的模型，特别是在真实世界研究中[43–45]。随着数据输入的增加，模型性能将变得更加稳定，这是mac的一个特征。

据我们所知，这是首个基于多中心真实世界证据（RWE）预测ZF诱导异常肝功能的模型，该模型通过根据不同的ALT水平进行分组预测，为个体提供及时的决策支持。涉及来自多家医院的患者 的多中心研究可以提高临床资源的利用率和大数据集的挖掘能力，从而使研究结果具有更好的普适性。该预测模型还在不同的医疗中心进行了外部验证，以验证模型的稳健性。此外，我们通过机器学习和深度学习技术探索了影响异常肝功能的重要变量，这些技术非常适合高维数据挖掘。第三，我们比较了不同的变量集以实现最佳预测性能，并最终获得了一个简明而准确的模型。最后，为了进一步应用，我们建立了一个易于使用的在线预测网站。

真实世界研究的局限性是显而易见的。由于这些研究是非实验性的，结果不可避免地受到了各种混杂因素的影响。由于来自不同中心的大量真实世界数据，每个研究的技术水平、设备状况和患者背景

Table 4
Model performance.

Model	ALT group	Precision	Recall rate	F1-score	AUC	Sensitivity	Specificity	Accuracy
LR	0	0.95	0.99	0.97	0.91	0.99	0.35	0.94
	1	0.33	0.11	0.17	0.86	0.11	0.99	
	2	0.80	0.50	0.62	0.99	0.50	1.00	
RF	0	0.95	1.00	0.98	0.98	1.00	0.31	0.95
	1	0.57	0.22	0.32	0.97	0.22	0.99	
	2	1.00	0.13	0.22	0.99	0.13	1.00	
ANN	0	0.95	0.99	0.97	0.95	0.99	0.31	0.93
	1	0.14	0.06	0.08	0.92	0.06	0.98	
	2	0.75	0.38	0.50	0.99	0.38	1.00	
SVM	0	0.94	1.00	0.97	0.95	1.00	0.12	0.94
	1	0.00	0.00	0.00	0.92	0.00	1.00	
	2	0.75	0.38	0.50	0.99	0.38	1.00	
KNN	0	0.96	0.97	0.97	0.90	0.97	0.42	0.92
	1	0.25	0.28	0.26	0.81	0.28	0.96	
	2	0.00	0.00	0.00	0.87	0.00	1.00	
CatBoost	0	0.98	0.97	0.98	0.98	0.97	0.69	0.95
	1	0.55	0.61	0.58	0.97	0.61	0.98	
	2	0.86	0.75	0.80	0.99	0.75	1.00	
LightGBM	0	0.99	0.97	0.98	0.98	0.97	0.85	0.96
	1	0.60	0.83	0.70	0.98	0.83	0.97	
	2	0.83	0.63	0.71	0.97	0.63	1.00	
XGBoost	0	0.95	1.00	0.97	0.98	1.00	0.23	0.94
	1	0.67	0.22	0.33	0.97	0.22	0.99	
	2	0.00	0.00	0.00	0.99	0.00	1.00	
GBDT	0	0.99	0.96	0.97	0.98	0.96	0.81	0.95
	1	0.50	0.78	0.61	0.97	0.78	0.96	
	2	0.71	0.63	0.67	0.97	0.63	0.99	
TabNet	0	0.96	0.99	0.97	0.94	0.99	0.38	0.93
	1	0.27	0.17	0.21	0.92	0.17	0.98	
	2	0.75	0.38	0.50	0.92	0.38	1.00	

Label “0” indicates patients with ALT < 40 U/L, Label “1” indicates patients with 40 ≤ ALT < 80 U/L, and Label “2” indicates patients with ALT ≥ 80 U/L. ALT: alanine transaminase; ANN: artificial neural network; AUC: area under the curve; CatBoost: categorical boosting; GBDT: gradient boosting decision tree; KNN: k-nearest neighbor; LightGBM: light gradient boosting machine; LR: logistic regression; RF: random forest; SVM: support vector machine; XGBoost: extreme gradient boosting.

in this study was more beneficial for elderly patients and middle-aged female patients. The model predictions allow patients to be aware of potential risks to their liver function when using ZF to help manage RA.

This study compared 10 machine learning and deep learning algorithms. After comparing the predictive metrics, LightGBM was chosen for modeling the risk of developing abnormal liver function. LightGBM is an improved version of the GBDT framework that involves fast, distributed and high-performance methods based on decision trees [13,42]. LightGBM uses three techniques, gradient-based one-side sampling, exclusive feature bundling, and histogram analysis, to require fewer samples, fewer features, and less memory [13,42]. LightGBM has been optimized for parallel computing, supporting feature parallelism and data parallelism, and has been optimized for each parallel mode to reduce traffic [13,42]. In clinical applications, machine learning and deep learning techniques can be applied to develop models with strong generalizability and good robustness, especially in real-world studies [43–45]. With increasing data input, the model performance will become more stable, which is a characteristic of machine learning and deep learning techniques. In short, the LightGBM can build a robust and accurate model relying on fewer samples and less traffic that can be used well for prediction and analysis of large datasets.

To our knowledge, this is the first model for predicting ZF-induced abnormal liver function based on multicenter RWE, which involves grouping predictions according to different ALT levels to provide prompt decision support for individuals. Multicenter studies involving patients from numerous hospitals can increase the utilization of clinical resources and the ability to mine large datasets, resulting in better generalization of the study outcomes. The prediction model was also externally validated in a different medical center to verify the model robustness. Furthermore, we explored the important variables influencing abnormal liver function through machine learning and deep learning techniques, which are well-suited to high-dimensional data mining. Third, we compared different variable sets to achieve the best predictive performance and ultimately obtained a concise and accurate model. Lastly, for further application, we established an easy-to-use online prediction website for clinicians. In brief, we proposed a model to provide accurate prediction of ZF-induced abnormal liver function, which can be easily used in clinical practice for medication guidance.

The limitations of real-world studies are evident. Since these studies were nonexperimental, the outcomes were inevitably influenced by various confounding factors. Due to the large amount of real-world data from different centers, the technical levels, equipment conditions and patient backgrounds at each research

RA-Zhengqing Fengtongning-Abnormal liver function

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Please input sample data in the text form:

Age (≥18)

68

Gender-Age Group

elderly male

▼

Smoking?

Yes

▼

Total cholesterol (mmol/L)

7.01

Triglyceride (mmol/L)

1.18

Treatment Plan

MTX+ZF

▼

Reset

Submit

Prediction result:

This sample is identified for predicting ALT≥80 U/L

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图5. 预测ZF引起的肝功能异常的网页应用程序。可通过以下网站访问该网页：https://119.3.220.198/work/pec1.html。ALT：丙氨酸转氨酶；MTX：甲氨蝶呤；RA：类风湿性关节炎；ZF：正清风痛宁。

中心可能会引入偏差和多样性，从而影响临床数据的同质性。在此，本研究涉及的所有医院均遵守《医疗机构检查检验结果互认管理办法》，确保严格的质量控制标准。这些标准符合国家级质量评估标准，有利于医院之间检测结果的互认。此外，一些医疗记录数据，如尿酸、白蛋白、低密度脂蛋白胆固醇和肌酐水平，缺失率较高，主要由于记录遗漏，这可能影响最终结果。同时，一些变量，如肝相关疾病、代谢紊乱、治疗过程、其他药物使用情况以及AST水平未被纳入。我们期望加强医务人员在医疗记录适当维护方面的培训，以在随访中纳入更全面的数据。

5. 结论

本研究使用ZF为RA患者异常肝功能构建了一个新的预测模型，并为安全用药提供指导。对真实世界数据进行了详细探索。在数据挖掘和建模过程中引入了机器学习和深度学习技术。在我们测试的10种算法中，LightGBM在预测指标和模型稳定性方面表现优越。该模型可能帮助临床医生在临床实践中实时预测和预防ZF引起的异常肝功能的发生。

CRedit 作者贡献声明

ZY 设计了这项研究，分析了数据，并撰写了稿件初稿。YG 和 FK 收集了数据。HW 提供了医疗指导。FG 提供了方法学指导。CL 提供了药理学指导。所有作者都审阅了稿件。

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利益冲突声明

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附录 A. 补充材料

本文的补充材料可在 <https://doi.org/10.1016/j.joim.2024.12.001> 在线获取。

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68

Gender-Age Group

elderly male

▼

Smoking?

Yes

▼

Total cholesterol (mmol/L)

7.01

Triglyceride (mmol/L)

1.18

Treatment Plan

MTX+ZF

▼

Reset

Submit

Prediction result:

This sample is identified for predicting ALT≥80 U/L

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Fig. 5. The web application for predicting ZF-induced abnormal liver function. The web can be accessed through the following website: https://119.3.220.198/work/pec1.html. ALT: alanine transaminase; MTX: methotrexate; RA: rheumatoid arthritis; ZF: Zhengqing Fengtongning.

center may introduce bias and diversity, thus affecting the homogeneity of the clinical data. Herein, all hospitals involved in this study adhere to the “Management Measures for the Mutual Recognition of Medical Institution Examination and Inspection Results,” ensuring rigorous quality control standards. These standards are in line with national-level quality assessment criteria, facilitating the mutual recognition of test results among hospitals. In addition, some medical records data, such as uric acid, albumin, low-density lipoprotein cholesterol and creatinine levels, were missing at high rates, mostly due to recording omittance, which may have affected the final outcome. Also, some variables, such as liver-related diseases, metabolic disorders, course of treatment, use of other drugs, and AST levels were not included. We expect to enhance the training of medical staff in the appropriate maintenance of medical records in order to include more comprehensive data in follow-up studies. In the future, a prospective study should be conducted to provide evidence that supports our findings.

5. Conclusion

This study prepared a new prediction model for abnormal liver function in RA patients using ZF and provides guidance for safe drug use. Real-world data were explored in detail. Machine learning and deep learning techniques were introduced in the process of data mining and modeling. Among the 10 algorithms we tested, the LightGBM exhibited superior performance in terms of prediction metrics and model stability. This model may help clinicians predict and prevent the occurrence of abnormal liver function caused by ZF in real-time in clinical practice.

CRedit authorship contribution statement

ZY designed the study, analyzed data, and wrote the draft manuscript. YG and FK collected data. HW provided medical guidance. FG provided methodological guidance. CL provided pharmacological guidance. All authors reviewed the manuscript.

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Declaration of competing interests

The authors declare the following personal relationships, which may be considered potential competing interests: Fei Gao was employed by Beijing Medicinovo Technology Co., Ltd.

Appendix A. Supplementary material

Supplementary material to this article can be found online at <https://doi.org/10.1016/j.joim.2024.12.001>.

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